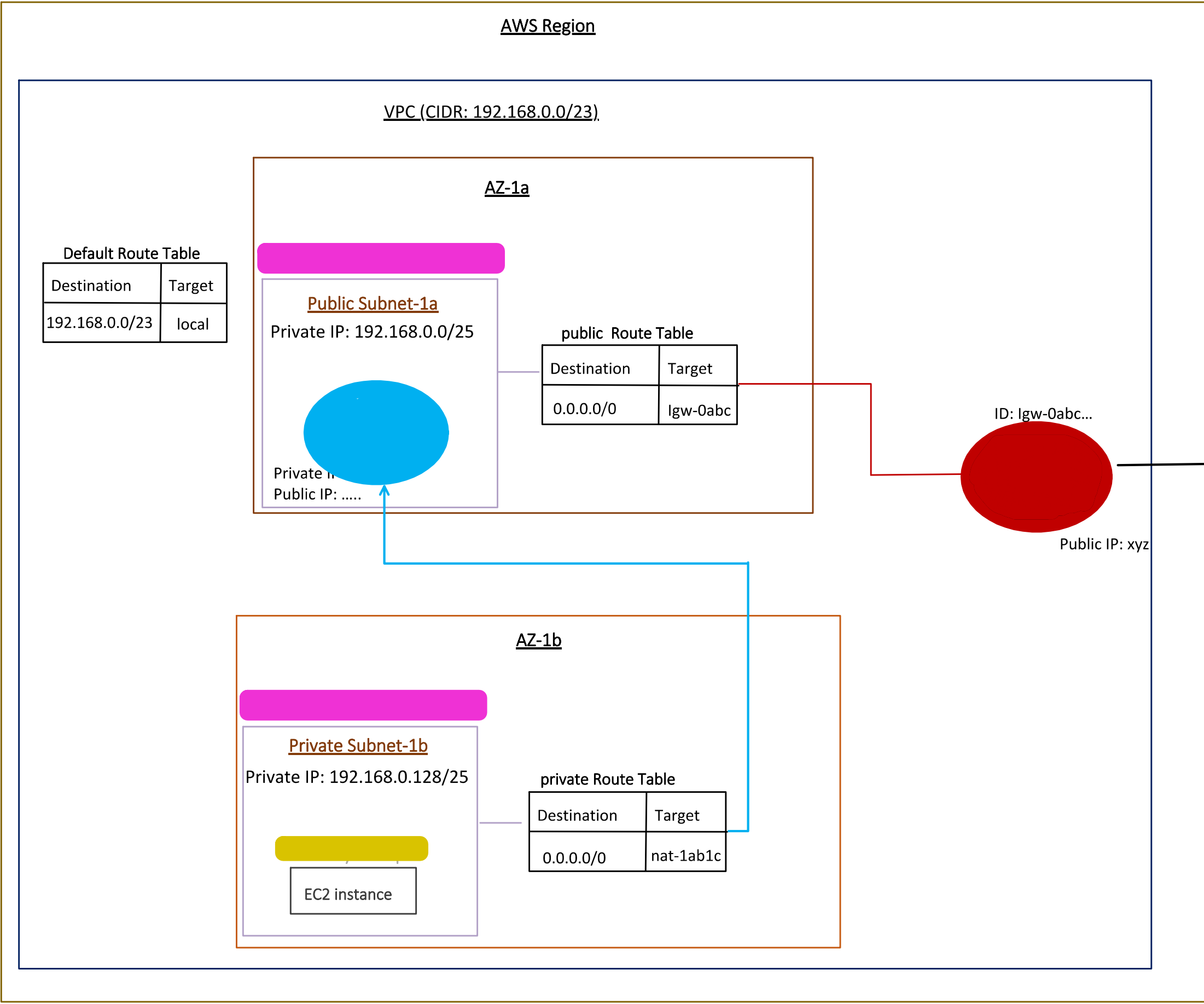
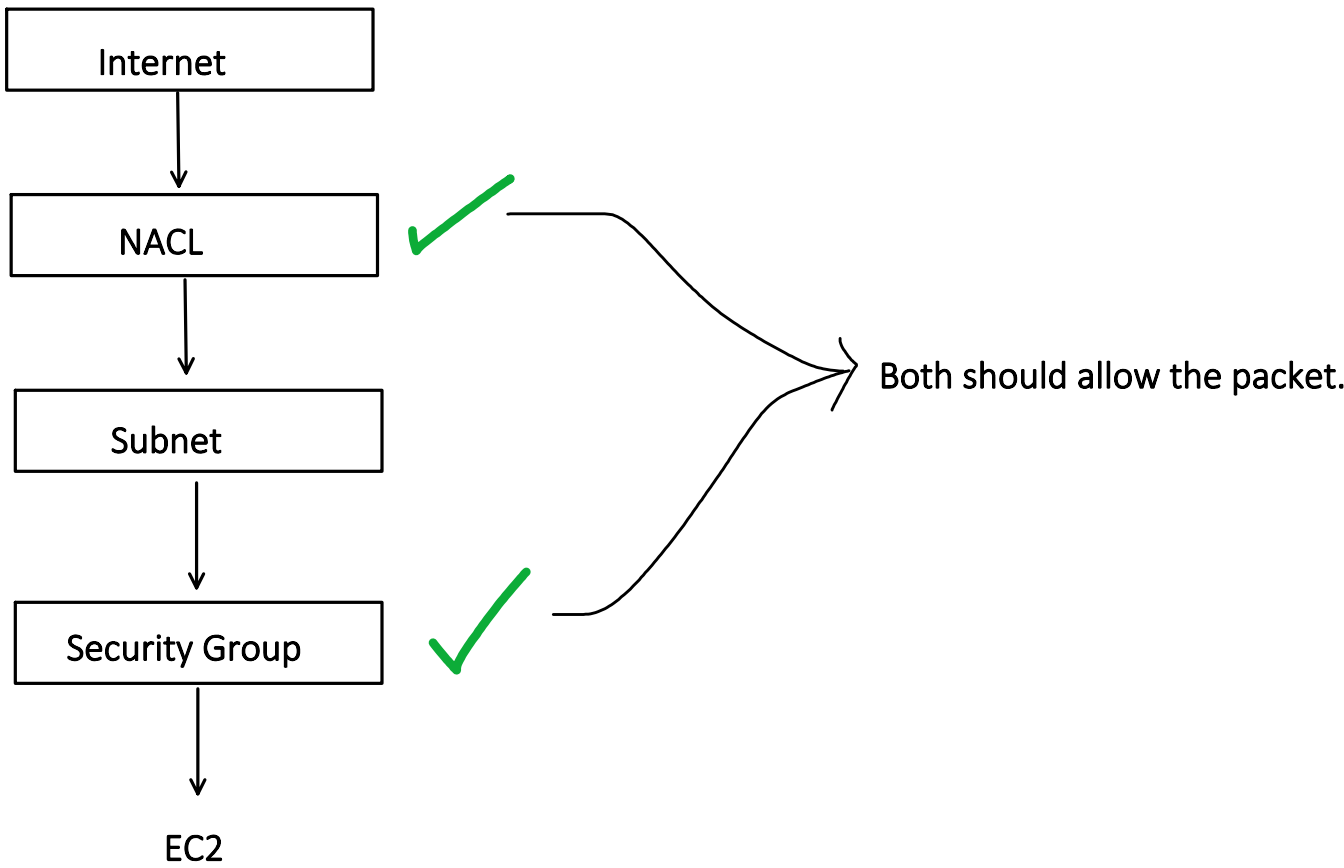


Security Group and NACL

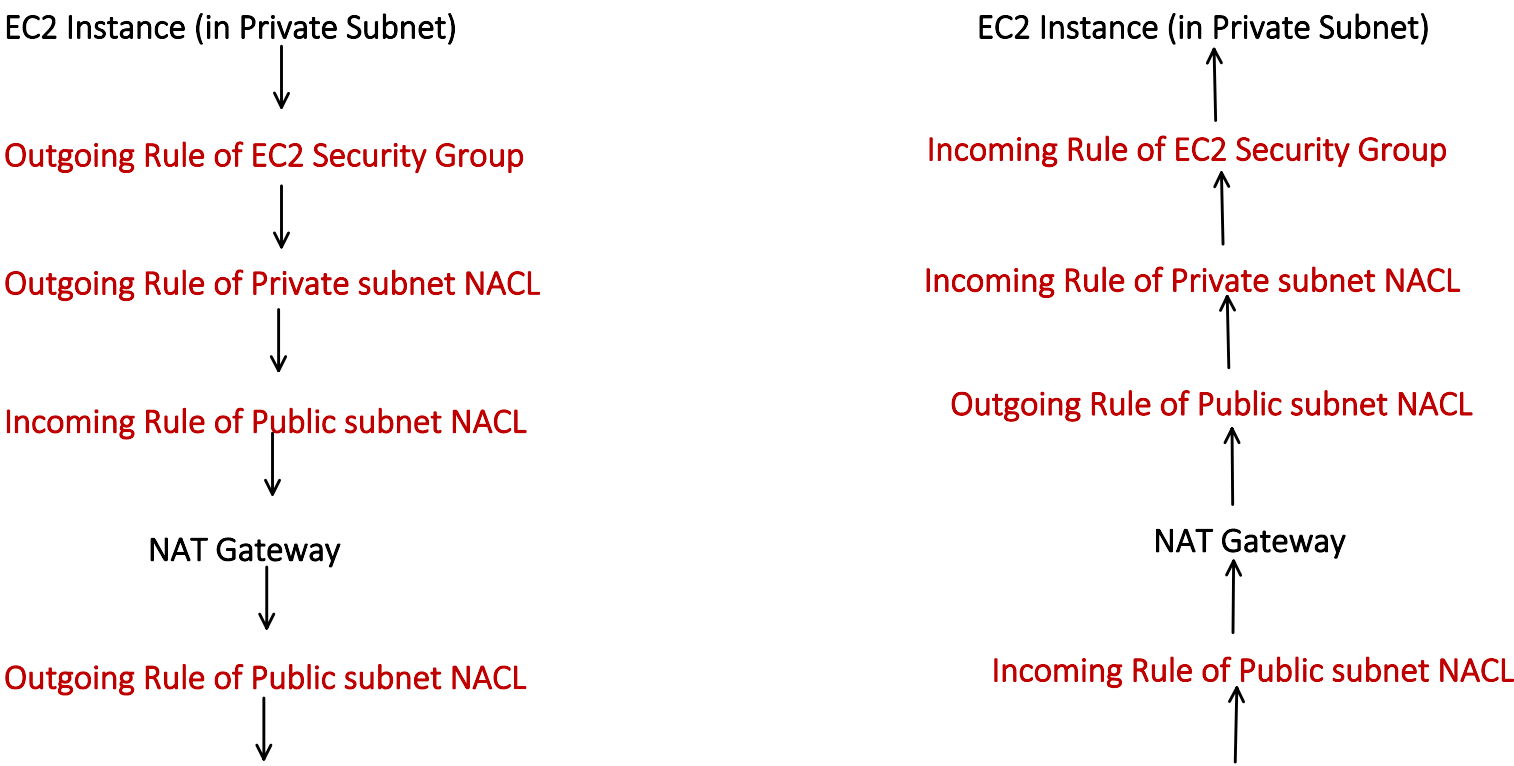


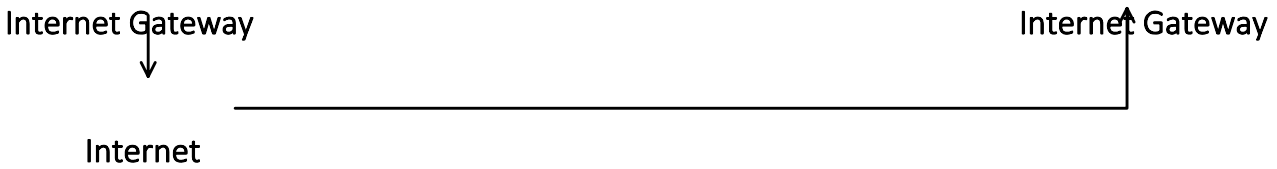
- NACL (Network Access Control List);
- Incoming and Outgoing Rules applied at SUBNET Level
 - Like a security guard outside the building
 - Stateless
 - means, must explicitly set rules for both inbound and outbound.
 - Support both Allow and Deny Rules
 - It has Rule numbers, lower the rule, higher the priority. If no rule match, traffic is denied by implicit rule.

- Security Group
- Incoming and Outgoing Rules applied at host Level like EC2, RDS, ALB etc.
 - Like a security guard outside the particular door.
 - Stateful : replies are auto allowed
 - means:
Outbound allowed : Allow HTTPS (443)
EC2 --> Google then response from Google ----> EC2 is also allowed no Inbound rule required for it.
 - similarly:
Inbound allowed: Allow HTTPS (443)
User ----> EC2, then response from EC2 ----> User is also allowed no Outbound rule is needed.
 - By default all inbound and outbound traffic is denied. So we only need to set ALLOW rules



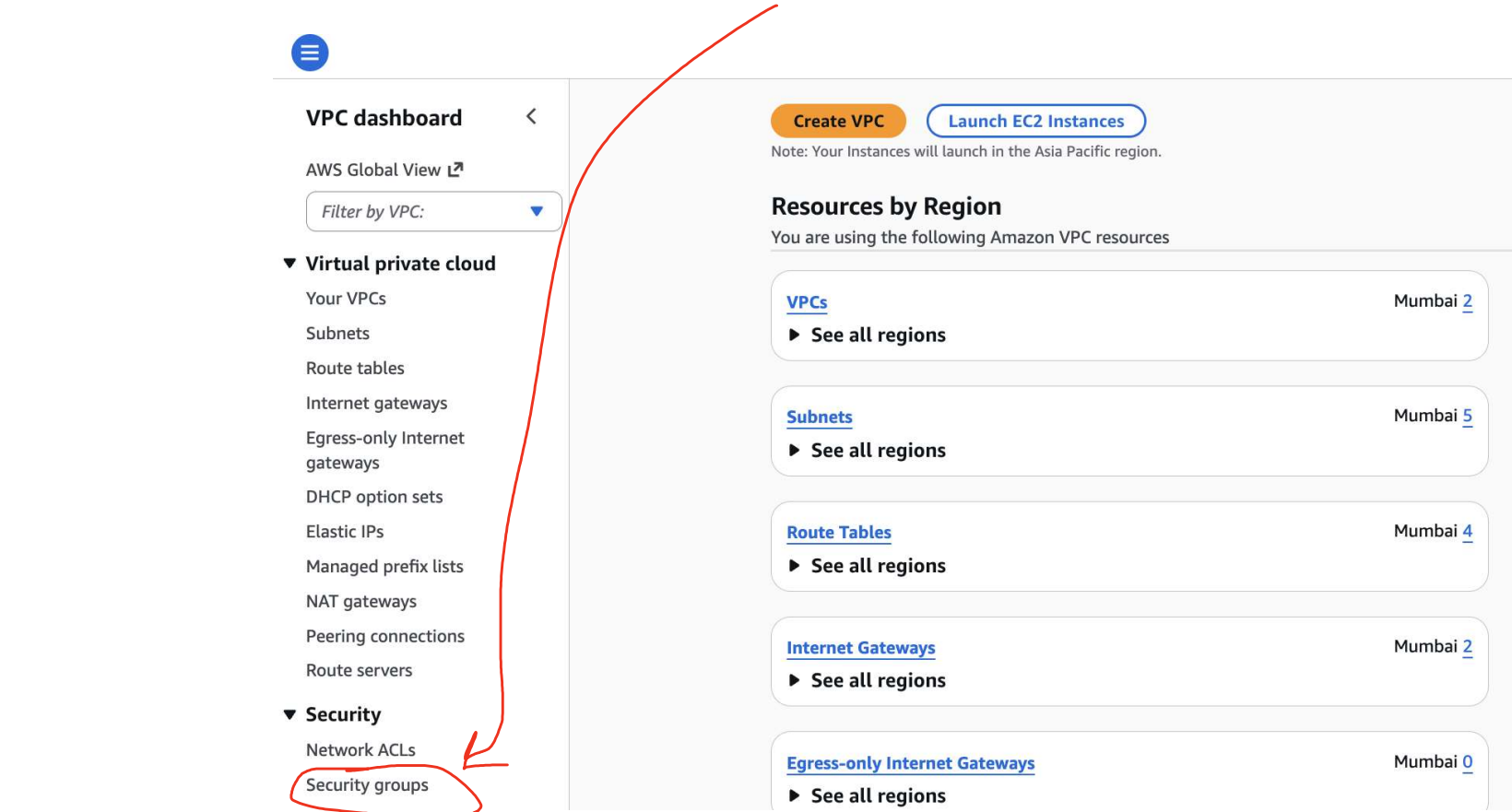
traffic flow from future EC2 instance in private subnet:





Security Group setup:

VPC -> Security Group



Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name

private-sg-ec2

Name cannot be edited after creation.

Description

private ec2, no inbound

VPC

vpc-0e288bbc758779d85 (vpc-conceptsbyshtayansh-prod)

Inbound rules

This security group has no inbound rules.

Add rule

Outbound rules

Type

HTTPS

Protocol

TCP

Port range

443

Destination

Anywhere...

Description - optional

Add rule

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs. No tags associated with the resource.

Add new tag

Cancel

Create security group

Now during EC2 instance creation, we can select the Security group

EC2 > Instances > Launch an instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simp

Name and tags

Name

e.g. My Web Server

Add additional tags

Network settings

VPC - required

vpc-0e288bbc758779d85 (vpc-conceptsbyshtayansh-prod)

Select VPC

Subnet

subnet-06f64e033b2a4addc public-subnet-1a

Create new subnet

Auto-assign public IP

Enable

Firewall (security groups)

Create security group

Select existing security group

Now select the security group which we want to attach with EC2 instance.

Common security groups

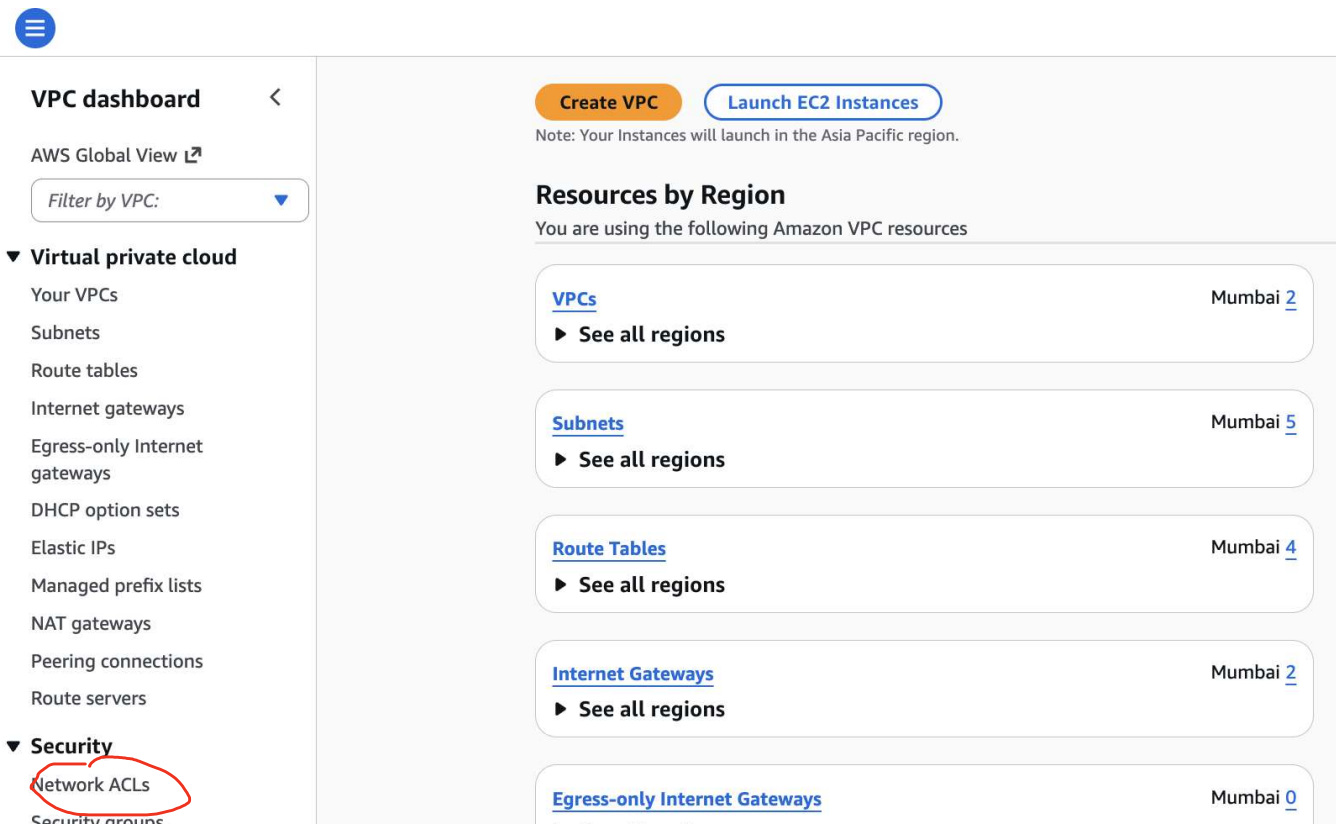
Select security groups

private-sg-ec2 sg-07e7f989dd895c2bf

Compare security group rules

Advanced network configuration

NACL Setup:



Set name -> VPC -> create NACL

Create network ACL info

A network ACL is an optional layer of security that acts as a firewall for controlling traffic in and out of a subnet.

Network ACL settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

private-subnet-NACL

VPC
VPC to use for this network ACL
vpc-0c288bbc758779d85 (vpc-conceptsbvshrayanah-prod)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name X

Value - optional

Q private-subnet-NACL X Remove tag

Add tag

You can add 40 more tags

Cancel Create network ACL

Set Inbound rules:

You successfully created acl-0f19affd74d4ee2a0 / private-subnet-NACL.

acl-0f19affd74d4ee2a0 / private-subnet-NACL Actions

Details info

Network ACL ID
acl-0f19affd74d4ee2a0

Owner
643537055131

Associated with
-

Default
No

VPC ID
vpc-0c288bbc758779d85 / vpc-conceptsbvshrayanah-prod

Inbound rules Outbound rules Subnet associations Tags

Inbound rules (1)

Filter inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
*	All traffic	All	All	0.0.0.0/0	Deny

Edit inbound rules

Allowing all traffic

Edit inbound rules info

Inbound rules control the incoming traffic that's allowed to reach the VPC.

Rule number info

100

Type info

All traffic

Protocol info

All

Port range info

All

Source info

0.0.0.0/0

Allow/Deny info

Allow

Remove

*

All traffic

All

All

0.0.0.0/0

Deny

Add new rule Sort by rule number

Cancel Preview changes Save changes

Set outbound rules:

acl-0f19affd74d4ee2a0 / private-subnet-NACL Actions

Details info

Network ACL ID
acl-0f19affd74d4ee2a0

Owner
643537055131

Associated with
-

Default
No

VPC ID
vpc-0c288bbc758779d85 / vpc-conceptsbvshrayanah-prod

Inbound rules Outbound rules Subnet associations Tags

Outbound rules (1)

Filter outbound rules

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
*	All traffic	All	All	0.0.0.0/0	Deny

Edit outbound rules

Allowing all traffic

Edit outbound rules info

Outbound rules control the outgoing traffic that's allowed to leave the VPC.

Rule number info

200

Type info

All traffic

Protocol info

All

Port range info

All

Destination info

0.0.0.0/0

Allow/Deny info

Allow

Remove

*

All traffic

All

All

0.0.0.0/0

Deny

Add new rule Sort by rule number

Cancel Preview changes Save changes

Now associate this NACL with Subnet:

acl-00683e4ac64f10b5d

acl-00683e4ac64f10b5d Actions

Details info

Network ACL ID
acl-00683e4ac64f10b5d

Owner
643537055131

Associated with
2 Subnets

Default
Yes

VPC ID
vpc-0c288bbc758779d85 / vpc-conceptsbvshrayanah-prod

Inbound rules Outbound rules Subnet associations Tags

Outbound rules (2)

Filter outbound rules

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Edit outbound rules

Edit subnet associations info

Change which subnets are associated with this network ACL.

Available subnets (1/2)

Filter subnet associations

Name	Subnet ID	Associated with	Availability Zone	IPv4 CIDR	IPv6 CIDR
☐ public-subnet-1a	subnet-06f64a03b2a4addc	acl-00683e4ac64f10b5d	ap-s1-az1 (ap-south-1a)	192.168.0.0/25	-
☒ private-subnet-1b	subnet-0c36555e6d75397d8	acl-00683e4ac64f10b5d	ap-s1-az3 (ap-south-1b)	192.168.0.128/25	-

Selected subnets

subnet-0c36555e6d75397d8 / private-subnet-1b X

Cancel Save changes

